EGA: Adapting Frozen Encoders for Vector Search with Bounded OOD Degradation

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Abstract

Vector similarity search systems built on frozen vision-language encoders routinely face queries from classes absent at adapter training time, since both the indexed database and the live query stream contain categories that were never labeled. We study how lightweight adapters behave under this distribution shift, and find a consistent in-distribution/out-of-distribution (ID/OOD) tradeoff among existing training paradigms. Global contrastive objectives such as ICon and SRL, when paired with high-capacity adapters, achieve near-perfect in-distribution Label Precision on CIFAR-100 (0.999 and 0.992 at K=1, nprobe=1) but collapse to 0.562 and 0.552 on the unseen classes of Food-101, a drop of 44 percentage points. Low-capacity LoRA variants avoid this collapse but underperform substantially in-distribution. We propose Euclidean Geodesic Alignment (EGA), a residual adapter trained with a small-margin triplet loss on the class-aware neighborhood graph of pre-trained embeddings. EGA induces gradient sparsity at convergence (96.5% of triplets produce zero gradient), empirically leaving the unseen-class geometry largely intact while permitting high-capacity in-distribution refinement. Across CIFAR-100 (ID) and three OOD benchmarks (CIFAR-10, FGVC-Aircraft, Food-101), EGA reaches the highest in-distribution Label Precision among adapters that avoid OOD collapse (0.705 versus 0.6176 for capacity-matched LoRA at r=128), with bounded OOD degradation (-7 to -9 pp, versus -32 to -33 pp for ICon and SRL). The benefits transfer to stronger backbones: on DINOv2-large and SigLIP, EGA improves Label Precision@1 by 4.75 and 8.55 percentage points.

1 Introduction

A vector similarity search system deployed in production rarely operates on a fixed label space. A retailer indexes product images today and adds a new category line next month; a content platform builds an index over existing tags and continually ingests media from emerging topics; a scientific image database is annotated for known phenotypes and queried for ones discovered later. In such scenarios, the index and the query stream together cover a label space strictly larger than what was available when any model was trained or adapted.

This deployment reality sits awkwardly with how adapters over frozen vision-language encoders are currently trained. The dominant paradigm takes a labeled subset, optimizes the embedding geometry to separate those classes, and ships the resulting adapter as a drop-in replacement for the frozen encoder's output. The implicit assumption is that geometry tightened for seen classes will, at worst, leave unseen-class geometry unchanged. We show this assumption fails systematically: high-capacity adapters trained with global contrastive objectives reshape the entire embedding distribution, pulling unseen-class samples toward whichever seen-class clusters they happen to resemble in the pre-trained metric. The retrieval system inherits this distortion silently, since approximate nearest neighbor indices remain well-behaved geometrically while the neighbors they return belong to the wrong semantic class.

Two design choices in current adapter training are responsible. First, the loss function: global contrastive objectives such as ICon [1] and SRL [4] compute gradients over the full pairwise similarity matrix in every step, so seen-class supervision exerts pressure across the entire embedding distribution rather than only on labeled regions. Second, the adapter capacity: a sufficiently expressive adapter has the representational room to comply with that pressure, and zero-initialized residual designs do not by themselves prevent the cumulative perturbation from reaching unseen-class regions. Existing alternatives address one factor at a time. Low-rank adapters such as LoRA [10] cap the second factor, which avoids the worst out-of-distribution (OOD) failures but limits in-distribution (ID) refinement, while keeping the global loss leaves the first factor untouched. We argue that adapters intended to operate under unseen-class queries should be designed against both factors at once.

The cost of failing on either factor is large. On CIFAR-100, high-capacity global contrastive adapters achieve near-perfect Label Precision: ICon reaches 0.999 and SRL reaches 0.992 at $K = 1$, nprobe $= 1$. On the strictly unseen classes of Food-101, the same adapters collapse to 0.562 and 0.552, which is 44 percentage points below their in-distribution scores and between 32 and 33 percentage points below the frozen CLIP baseline. LoRA variants avoid this collapse but pay for it in-distribution, with Label Precision bounded between 0.61 and 0.67, well below what a high-capacity adapter can achieve when the loss does not over-extend into unseen-class regions.

We propose Euclidean Geodesic Alignment (EGA), a residual adapter designed against both factors simultaneously. EGA pairs a high-capacity zero-initialized residual architecture with a small-margin triplet loss on the class-aware neighborhood graph of pre-trained embeddings, with ℓ_2 projection onto the unit hypersphere. The triplet loss induces a self-limiting training dynamic: as the adapter converges on seen classes, the fraction of margin-violating triplets decays rapidly, reaching 96.5% zero-gradient triplets at steady state. The capacity to refine seen-class neighborhoods is preserved, but the loss stops applying pressure once seen-class neighborhoods are locally separated, so the global reshaping that destroys unseen-class geometry under ICon and SRL never accumulates.

We evaluate EGA on CIFAR-100 (ID) and three OOD benchmarks (CIFAR-10, FGVC-Aircraft, Food-101), comparing against the frozen CLIP baseline, global contrastive adapters (ICon, SRL), and LoRA configurations with ranks $r = 64$ and $r = 128$ that are capacity-matched to EGA. The six methods occupy distinct operating points in the (ID, OOD) Label Precision plane. EGA reaches the highest in-distribution Label Precision among adapters that avoid OOD collapse, achieving 0.705 on CIFAR-100 versus 0.6176 for capacity-matched LoRA at $r = 128$, while bounding OOD degradation between -7 and -9 percentage points on CIFAR-10 and Food-101 (compared to between -32 and -33 for ICon and SRL) and improving by 9.9 percentage points on FGVC-Aircraft. The benefits transfer to stronger backbones, with EGA improving Label Precision@1 by 4.75 and 8.55 percentage points on DINOv2-large and SigLIP. EGA does not dominate every method on every axis: on CIFAR-10 and Food-101, capacity-matched LoRA achieves higher OOD Label Precision, and deployments dominated by OOD queries on these distributions may prefer LoRA at the cost of weaker in-distribution refinement.

In summary, this work makes the following contributions:

- We frame adapter tuning over frozen vision-language encoders from a vector search deployment perspective, in which the index and query stream together cover a label space larger than the labeled subset available for adapter training, and we characterize the failure modes of existing adapter paradigms under this lens.
- We propose Euclidean Geodesic Alignment (EGA), a residual adapter that combines a high-capacity zero-initialized architecture with a small-margin triplet objective. The triplet loss converges to a sparse-gradient regime (96.5% zero-gradient triplets at steady state), which empirically preserves unseen-class geometry while permitting in-distribution refinement.
- Across CIFAR-100 and three OOD benchmarks, and across three backbones (CLIP ViT-B/32, DINOv2-large, SigLIP), EGA occupies a distinct operating point in the (ID, OOD) Label Precision plane, achieving the highest in-distribution Label Precision among adapters that avoid OOD collapse, while capacity-matched LoRA variants preserve OOD precision better on two of three benchmarks at the cost of 8.7 percentage points in-distribution.

2 Background and Related Work

Foundation models such as CLIP [25] produce embedding spaces that encode both seen-class semantics and broader priors over unseen visual concepts. Related work has documented degradation phenomena in adapted representations [20, 37], and recent work documents analogous degradation in contrastive representations, including anisotropic collapse [43, 6], modality gap fracturing [26, 5], and loss of compositional structure [24, 28, 36]. Recent strategies counter these issues with geometric constraints such as orthogonality regularization [27]. *EGA exploits the gradient sparsity of small-margin triplet losses to keep adapter updates concentrated on seen-class neighborhoods, empirically leaving unseen-class regions of the pretrained geometry largely intact.*

Recent representation learning frequently relies on global contrastive objectives to optimize the latent space [31, 19]. A common paradigm jointly enforces alignment for positive pairs and uniformity across the feature distribution [34], applied in graph encoders [33] and multimodal recommendation [46]. While uniformity improves robustness to noise [35], it treats all latent regions equally. Other structural regularizers include unifying frameworks for multi-view alignment [1], geometric constraints for imbalanced regression [4], decoupled objectives [18], information-theoretic density control [42], and region-based distillation [15]. *EGA replaces dense global optimization with sparse local triplet updates, which empirically reduces unseen-class distortion.*

Parameter-efficient adaptation is an approach for transferring large foundation models without full fine-tuning [9, 10], which dominates vision-language research [21]. Lightweight methods, e.g., Visual Prompt Tuning [14] and Tip Adapter [41], specialize the model without modifying the frozen backbone and rival full fine-tuning on many tasks [29], while zero-initialization strategies further constrain the magnitude of pretrained-manifold perturbation [40]. *EGA admits a zero-initialized residual adapter, localized triplet objective, and parameter efficiency with gradient-sparse updates.*

Vector search is the foundational engine for scalable high-dimensional retrieval. The systems community has developed indexing algorithms for billion-scale datasets, including FAISS [16], HNSW [22], DiskANN [30], and SPANN [2]. Recent work refines graph topologies [38, 3], optimizes SSD-based search [7, 8], and accelerates distance computations [13], yet popular ANN implementations still suffer worst-case limitations [12]. *EGA recalibrates the metric geometry before index construction, rendering search backends achieve higher semantic recall without algorithmic changes.*

3 Methodology

3.1 Problem Formulation

Let $\phi : \mathcal{X} \to \mathbb{R}^d$ denote a frozen vision-language encoder (e.g., CLIP ViT-B/32) that maps an image $x \in \mathcal{X}$ to a unit-normalized embedding $\mathbf{z} = \phi(x) \in \mathbb{S}^{d-1}$. We are given a set of labeled training images $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{N}$, where $y_i \in \mathcal{Y}_{\text{seen}}$ belongs to a set of *seen* classes. At inference time, the retrieval system must operate over a database that also contains images from a disjoint set of *unseen* classes $\mathcal{Y}_{\text{unseen}}$, with $\mathcal{Y}_{\text{seen}} \cap \mathcal{Y}_{\text{unseen}} = \emptyset$. Our goal is to learn an adapter $f_\theta : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ such that the transformed embeddings $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ support more accurate approximate nearest neighbor (ANN) retrieval on *both* seen and unseen classes, without access to unseen-class labels during training.

We evaluate retrieval quality along two complementary dimensions. *Label Precision@K* (LP@K) measures the semantic quality of retrieved results: whether the K nearest neighbors in the transformed space share the same class label as the query:

$$\text{LP@}K = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{K} \sum_{k=1}^{K} \mathbf{1}\left[y_{\hat{n}_k(q)} = y_q\right], \tag{1}$$

where $\mathcal{Q}$ is the query set and $\hat{n}_k(q)$ denotes the k-th nearest neighbor of q in the transformed embedding space. *ANNS Recall@K* (AR@K) measures the geometric fidelity of the ANN index, i.e, the fraction of true K nearest neighbors (under exact search) that are recovered by the approximate index at a given search budget nprobe:

$$\text{AR@}K = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{\left|\mathcal{N}_K^{\text{exact}}(q) \cap \mathcal{N}_K^{\text{approx}}(q)\right|}{K}, \tag{2}$$

where $\mathcal{N}_K^{\text{exact}}(q)$ and $\mathcal{N}_K^{\text{approx}}(q)$ denote the exact and approximate top-K neighbor sets, respectively. The two metrics capture distinct failure modes: an adapter may improve LP@K by tightening within-class clusters (semantic improvement) while simultaneously improving AR@K by making the geometry more amenable to IVF-based indexing (geometric improvement).

3.2 Manifold-Preserving Adapter Design

EGA introduces a lightweight adapter f_θ composed of stacked residual blocks with a final refinement layer. The architecture is constrained by three design principles, each of which contributes to keeping the adapter's updates anchored to the pretrained geometry.

(1) Residual connection with zero initialization Each EGA block applies a residual update as:

$$\mathbf{z}' = \mathbf{z} + g_\theta(\mathbf{z}), \tag{3}$$

where g_θ is a learned transformation initialized so that $g_\theta(\mathbf{z}) \approx \mathbf{0}$ at the start of training. Concretely, the final linear layer of g_θ is initialized to zero, making f_θ an identity mapping at initialization. This guarantees that training begins from the pretrained geometry rather than a random perturbation, and that the adapter can only *refine* rather than replace the CLIP embedding. As confirmed by our ablation (Table 4), removing the residual connection causes accuracy to collapse from 0.70 to 0.01, because the lightweight MLP cannot reconstruct complex semantic relationships from seen-class data alone.

(2) Feature gating Inside each residual block, the transformed signal passes through a module:

$$\text{Gate}(\mathbf{h}) = \mathbf{h} \odot \sigma(W_2\, \text{GELU}(W_1 \mathbf{h}))\,, \tag{4}$$

where $W_1 \in \mathbb{R}^{(d/4)\times d}$, $W_2 \in \mathbb{R}^{d\times(d/4)}$, σ is the Sigmoid function, and $\odot$ denotes elementwise multiplication. This is analogous to Squeeze-and-Excitation networks [11] to learn a per-dimension weight that amplifies retrieval-relevant dimensions and suppresses noisy ones. The bottleneck ratio $1/4$ is chosen to prevent the gating from overfitting on seen-class data, as validated in Table 5.

The output of f_θ is projected back onto the unit hypersphere $\mathbb{S}^{d-1}$ via ℓ_2:

$$\hat{\mathbf{z}} = \frac{f_\theta(\mathbf{z})}{\|f_\theta(\mathbf{z})\|_2}. \tag{5}$$

This constraint ensures that the transformed embeddings remain in the same metric space as the original CLIP embeddings, so that Euclidean distance and cosine similarity remain interchangeable on the hypersphere. Removing this constraint degrades retrieval performance by 4.2 percentage points (Table 4), confirming that hypersphere consistency is necessary for stable ANN indexing.

(3) Full forward pass Let Block_i denote the i-th EGA block consisting of a two-layer MLP with LayerNorm followed by feature gating, and let Refine denote the final linear projection with LayerNorm. The complete forward pass for an input embedding $\mathbf{z} \in \mathbb{S}^{d-1}$ is:

$$\hat{\mathbf{z}} = \ell_2(\text{Refine}(\text{Block}_2(\text{Block}_1(\mathbf{z}))))\,. \tag{6}$$

The total number of trainable parameters is approximately $4.2M$ for $d = 512$ and hidden dimension 2048, which is negligible compared to the frozen CLIP backbone (∼88M parameters).

3.3 Local Triplet Training

Given a mini-batch $\mathcal{B}$ sampled from $\mathcal{D}_{\text{train}}$, we construct triplets (a, p, n) where a is an anchor image, p is a positive sampled uniformly from the same class as a, and n is a negative sampled uniformly from a different class. The adapter is trained with the standard triplet margin loss:

$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{(a,p,n)\in\mathcal{B}} \max(0,\ d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m)\,, \tag{7}$$

where $d(\cdot,\cdot)$ denotes distance, $m > 0$ is the margin, and $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ is the transformed embedding.

A triplet (a, p, n) contributes a non-zero gradient to θ if and only if the margin constraint is *violated*:

$$d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m > 0. \tag{8}$$

We refer to such triplets as *active*. As the adapter converges on seen classes, increasingly many triplets satisfy the margin constraint and become inactive, contributing zero gradient. This sparsity has a direct consequence for out-of-distribution generalization. Because unseen classes are absent from $\mathcal{D}_{\text{train}}$, no triplet involving an unseen-class embedding is ever constructed. Provided that the adapter's updates remain effectively local (i.e., the cumulative perturbation induced by seen-class training gradients remains small in unseen-class regions of the embedding space), the manifold structure for unseen classes is largely preserved. The gradient sparsity mechanism enforces this locality: once seen-class neighborhoods satisfy the margin, the corresponding gradients vanish, and only the small fraction of actively-violated local relationships continue to be updated.

We formalize these intuitions in the following observation, which characterizes the steady-state behavior of triplet-trained adapters and its implications for manifold preservation.

Gradient sparsity at convergence. The triplet loss in Eq. (7) consists of hinge terms whose subgradient with respect to θ is exactly zero whenever the margin constraint is satisfied:

$$\nabla_\theta \ell_{apn} = \mathbf{1}[d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m > 0] \cdot \nabla_\theta \delta_{apn}, \tag{9}$$

where $\delta_{apn} = d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n)$. Define the *active triplet ratio* at training step t as

$$\rho(t) = \Pr_{(a,p,n)\sim\mathcal{B}} \Big[d(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_p^{(t)}) - d(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_n^{(t)}) + m > 0 \Big]. \tag{10}$$

Under the standard assumption that training converges to a stationary point of $\mathcal{L}$, $\rho(t)$ converges to a steady-state value $\rho^* \in [0, 1]$ determined by the margin m and the within-class distance distribution of the converged embeddings. We measure $\rho^* \approx 3.5\%$ at convergence on FGVC-Aircraft (Section 4.3), meaning that 96.5% of training triplets contribute zero gradient at steady state.

From sparsity to OOD locality: an empirical claim. Strictly speaking, sparse seen-class gradients do not formally imply that f_θ leaves unseen-class regions unchanged: the adapter is a parametric MLP, not a locally supported transformation, so any update to θ propagates globally. We therefore make the weaker, empirical claim that the residual zero-initialized architecture combined with gradient sparsity keeps the cumulative perturbation in unseen-class regions small enough to preserve their pretrained semantic structure for retrieval. The OOD measurements in Section 4.2 are the evidence: EGA preserves or improves Label Precision on three unseen-class benchmarks, while ICon and SRL lack the gradient sparsity and fall below the frozen baseline. A formal Lipschitz-based bound on the unseen-class perturbation is left for future work; we discuss technical conditions in Appendix A.

4 Evaluation

Experiments were conducted on a server with 256 AMD EPYC-7763 CPU cores, 512 GiB RAM, and an NVIDIA A100 GPU of 80 GB HBM2e. Evaluation datasets include CIFAR-100 for in-distribution scenarios, CIFAR-10 for cross-dataset OOD settings, and FGVC-Aircraft and Food-101 for unseen-class OOD evaluations. All of these datasets can be directly downloaded from the Pytorch library. The backbone models include ClIP ViT-B/32 [25], DINOv2-large [23], and SigLIP [39]. We compare our method against four adapter models: ICon [1], SRL [4], LoRA + InfoNCE [10, 32], and LoRA + Triplet [10]; We omit Linear Probing, Tip-Adapter [41], and prompt tuning methods (CoOp/CoCoOp [45, 44]) because they are not compatible with our cross-dataset OOD setting. Additional experimental setup details are in Appendix B; extended experimental results, including incomplete baselines, can be found in Appendix C.

4.1 In-Distribution Validation

Figure 1 reports Label Precision@K across four neighborhood sizes and three search budgets. EGA raises Label Precision@1 from 0.549 (raw CLIP) to 0.705 at nprobe $= 1$, and the gap persists across all K and nprobe values, including nprobe $= 10$, indicating that the adapter tightens within-class neighborhoods rather than benefiting from broader index coverage.

The Label Precision improvement has a direct geometric correlate. Figure 2 contrasts the L_2 distance distributions for true Top-K neighbors against random background points. In the raw CLIP space, the two distributions overlap substantially, which is the condition under which IVF-based ANN indexing

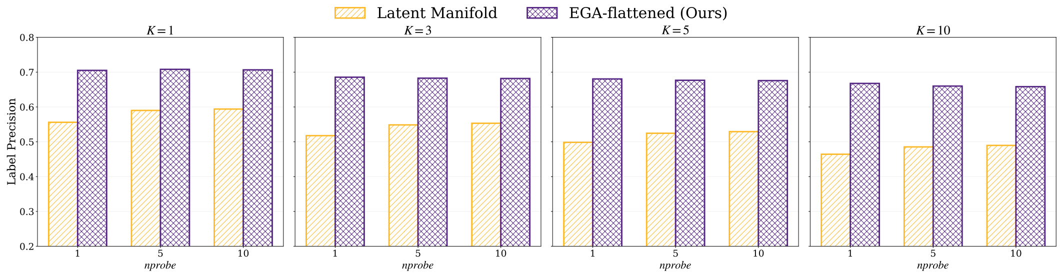

Figure 1: Label Precision@K on CIFAR-100 across $K \in \{1, 3, 5, 10\}$ and nprobe $\in \{1, 5, 10\}$.

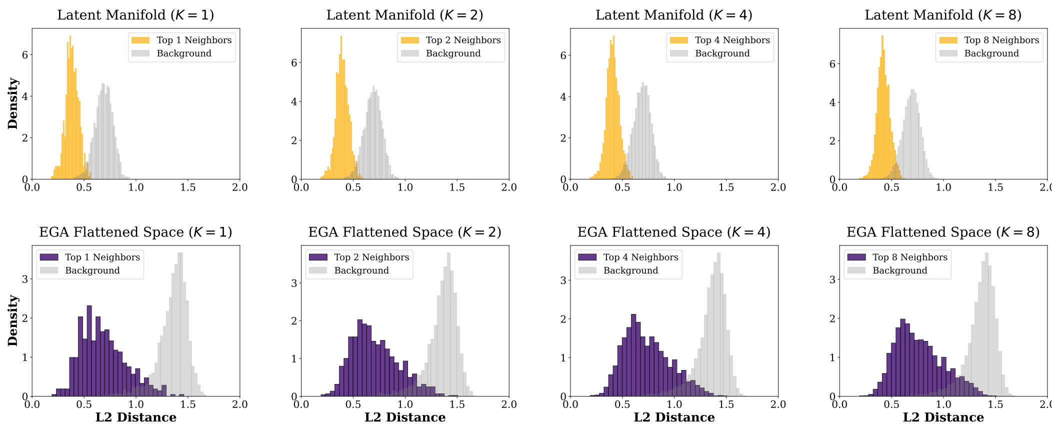

Figure 2: Distance distributions for Top-K neighbors vs. random background points on CIFAR-100.

Table 1: In-distribution retrieval performance on CIFAR-100 at $K = 1$, nprobe $= 1$. Standard errors are below 0.01 and omitted for clarity.

Metric	CLIP	ICon	SRL	LoRA+InfoNCE	LoRA+Triplet	EGA
LP@1	0.549	**0.999**	**0.992**	**0.668**	**0.612**	**0.705**
AR@1	0.667	**0.992**	**0.991**	**0.879**	**0.908**	**0.825**

degrades. After EGA, the two distributions separate cleanly across all K, and this separation drives the recall-efficiency improvement in Figure 3: raw CLIP achieves 0.667 Recall@1 at nprobe $= 1$, whereas EGA reaches 0.825 and saturates above 0.99 by nprobe $= 5$.

Table 1 reports all six methods' in-distribution performance, since the contrast with their out-of-distribution behavior in Section 4.2 is central to our analysis. ICon and SRL reach near-perfect Label Precision (0.999 and 0.992) by maximally tightening seen-class clusters under their global contrastive objectives. EGA reaches 0.705, an improvement of $+16$ pp over the frozen baseline, and the LoRA variants land between EGA and the baseline. As Section 4.2 will show, the same property that lets ICon and SRL dominate this metric is what causes their catastrophic out-of-distribution collapse.

4.2 Out-of-Distribution Generalization

Under strict OOD settings where evaluation categories are disjoint from training categories, the two retrieval metrics tell different stories. Table 2 reports headline numbers at $K = 1$, nprobe $= 1$. We evaluate on three benchmarks: CIFAR-10 (trained on CIFAR-100), FGVC-Aircraft (80 seen / 20 unseen classes), and Food-101 (80 seen / 21 unseen classes). For a fair comparison, all methods train identical lightweight adapters on top of frozen CLIP features and differ only in their loss functions. Robustness across search budgets (nprobe $\in \{1, 5, 10\}$) is reported in Appendix C.

On ANNS Recall, the adapters either match or exceed the frozen CLIP baseline on Aircraft and Food-101, indicating that adapter training succeeds at geometric reorganization. On Label Precision, the picture inverts. ICon and SRL fall *below* the frozen baseline on every dataset, with degradation

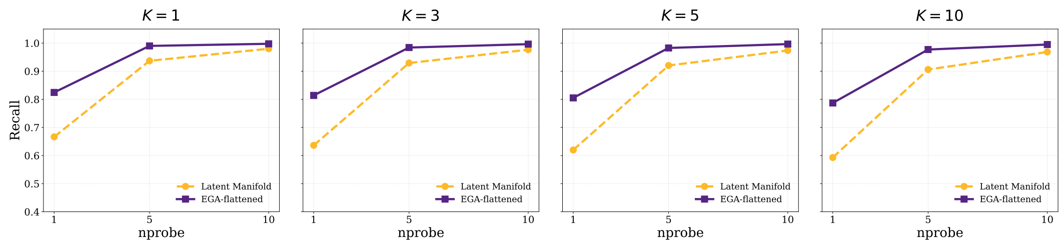

Figure 3: ANNS Recall@K versus nprobe on CIFAR-100.

Table 2: OOD performance at $K = 1$, nprobe $= 1$, mean $\pm$ std over 3 random seeds (42, 123, 456).

Dataset	Metric	CLIP	ICon	SRL	LoRA+InfoNCE	LoRA+Triplet	EGA
CIFAR-10	AR@1	0.863	0.797$\pm$.003	0.819$\pm$.001	**0.869**$\pm$.012	0.861$\pm$.011	0.833$\pm$.005
	LP@1	0.880	0.560$\pm$.010	0.536$\pm$.008	**0.885**$\pm$.006	0.875$\pm$.006	0.810$\pm$.002
Aircraft	AR@1	0.773	**0.853**$\pm$.016	**0.879**$\pm$.023	**0.891**$\pm$.015	**0.893**$\pm$.008	**0.905**$\pm$.015
	LP@1	0.512	0.470$\pm$.034	0.375$\pm$.032	**0.538**$\pm$.020	**0.569**$\pm$.011	**0.611**$\pm$.020
Food-101	AR@1	0.842	0.821$\pm$.001	0.824$\pm$.015	**0.906**$\pm$.003	**0.893**$\pm$.008	**0.879**$\pm$.011
	LP@1	0.881	0.562$\pm$.011	0.552$\pm$.010	0.883$\pm$.004	0.833$\pm$.007	0.791$\pm$.002

Table 3: Performance of EGA versus capacity-matched LoRA ($K = 1$, nprobe=1).

Dataset	Metric	EGA	InfoNCE r=64	InfoNCE r=128	Triplet r=64	Triplet r=128
CIFAR-100 (ID)	LP@1	**0.705**	0.6256	0.6176	0.5568	0.5472
	AR@1	**0.825**	0.879	0.879	0.908	0.908
Aircraft (OOD)	LP@1	**0.611**	0.5417	0.5298	0.5833	0.5417
	AR@1	**0.905**	0.6190	0.6131	0.6667	0.6726
Food-101 (OOD)	LP@1	0.791	**0.8667**	**0.8599**	**0.8599**	0.8568
	AR@1	**0.879**	0.6504	0.6702	0.6756	0.6740
CIFAR-10 (OOD)	LP@1	0.810	**0.8788**	**0.8728**	**0.8768**	0.8756
	AR@1	**0.833**	0.6008	0.5940	0.6216	0.6096

reaching -32 to -33 percentage points on Food-101 and -32 to -34 on CIFAR-10: the retrieved neighbors are geometrically closer in the IVF index but belong to the wrong semantic class. EGA improves Label Precision over the frozen baseline on Aircraft (+9.9 pp), with bounded degradation on CIFAR-10 (-7.0 pp) and Food-101 (-9.0 pp). EGA's advantage over global methods ranges from 14 pp on Aircraft to 23 pp on Food-101 and 25 pp on CIFAR-10, and tracks the difficulty of the OOD shift: when seen and unseen classes are visually similar, as in Food-101's fine-grained dishes, the dense gradient updates of ICon and SRL pull unseen samples into the wrong seen-class clusters.

Cross-referencing Table 1 and Table 2 reveals the structural tradeoff. ICon and SRL achieve near-perfect Label Precision on CIFAR-100 (0.999 and 0.992) but collapse to 0.562 and 0.552 on Food-101. The frozen baseline and LoRA variants sit at the opposite extreme, with modest in-distribution precision (0.55 to 0.67) but stable performance under distribution shift. EGA sits between these extremes: 0.705 in-distribution and 0.791 on Food-101, 0.611 on Aircraft, and 0.810 on CIFAR-10. The six methods occupy distinct operating points in the (ID, OOD) Label Precision plane, shown in Figure 4; the dashed line connects the points that are not strictly dominated. ICon and SRL achieve the highest in-distribution precision but collapse on OOD, while LoRA variants achieve competitive OOD precision on Food-101 and CIFAR-10 but trail EGA in-distribution by 8.7 percentage points.

Table 3 compares EGA against capacity-matched LoRA ($r = 64$ and $r = 128$). On CIFAR-100, EGA achieves 0.705 LP@1, exceeding the best capacity-matched LoRA by 7.9 percentage points. Across OOD benchmarks, EGA matches or exceeds LoRA on Aircraft and remains within 7–9 percentage points on Food-101 and CIFAR-10. LoRA's in-distribution performance slightly decreases with increasing rank, which we attribute to the low-rank constraint acting as an implicit regularizer.

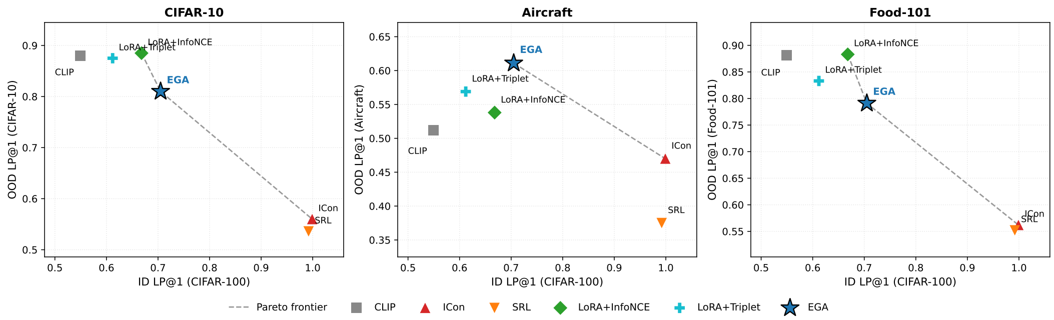

Figure 4: ID/OOD operating points of adapter training paradigms in the (ID, OOD) Label Precision plane. The dashed line connects the non-dominated points.

4.3 Two Routes to OOD Stability: Gradient Sparsity and Capacity Restriction

The introduction identified two factors through which adapter training can distort unseen-class geometry: a loss that exerts pressure across the full embedding distribution, and an adapter capacity expressive enough to comply with that pressure. Adapters that avoid OOD collapse must neutralize one of these factors, and the route determines its position in the Label Precision plane.

Route 1: Gradient sparsity (EGA). A triplet loss with margin m produces a non-zero gradient for a sample triplet (a, p, n) only when the margin constraint is violated:

$$d(a, p) - d(a, n) + m > 0. \tag{11}$$

As the adapter converges on seen classes, the fraction of *active* triplets decays. Figure 5 (left) tracks this ratio from 17.4% at epoch 10 to 3.5% at convergence. At steady state, 96.5% of triplets contribute zero gradient: the loss stops applying pressure once seen-class neighborhoods are locally separated, even though the adapter retains the capacity to refine them further. Sparsity neutralizes the loss-side factor while leaving capacity intact, which permits EGA's high in-distribution Label Precision in Table 1.

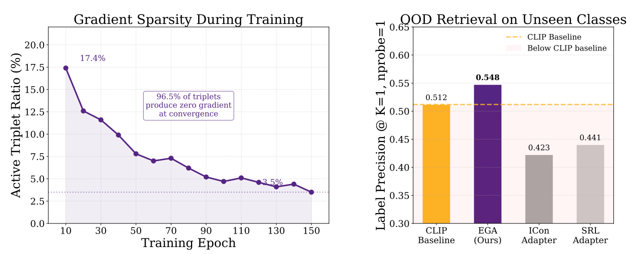

Figure 5: Gradient sparsity as OOD protection.

Route 2: Capacity restriction (LoRA). The LoRA variants take the other route. They retain a global contrastive loss (InfoNCE) or a triplet loss, but constrain the adapter to a low-rank residual update. Even at $r = 128$, LoRA's expressive power is limited to rank-128 perturbations of the identity, so the dense gradient field has substantially less room to reshape unseen-class geometry. This explains why LoRA+InfoNCE, despite using the same loss family as ICon, avoids the same OOD collapse: the loss applies pressure everywhere, but the adapter cannot fully comply. The price is paid in-distribution. At capacity-matched scale ($r = 128$, $\sim$4.2M parameters), LoRA+InfoNCE reaches 0.6176 Label Precision on CIFAR-100, 8.7 percentage points below EGA, because the same low-rank constraint that bounds OOD damage also bounds in-distribution refinement.

The two routes occupy different points in Figure 4. EGA, which neutralizes the loss-side factor while keeping capacity, achieves the highest in-distribution Label Precision among adapters that avoid OOD collapse. Capacity-matched LoRA, which neutralizes the capacity-side factor while keeping a global loss, achieves higher OOD Label Precision on CIFAR-10 and Food-101 at the cost of in-distribution refinement. Whether either route subsumes the other under stronger conditions, for example by combining gradient sparsity with low-rank constraints, is a question we leave open.

4.4 Ablation and Sensitivity Analysis

Table 4 isolates the contribution of each architectural component. The residual connection is the core element: removing it collapses accuracy from 0.7015 to 0.0065, indicating that the adapter cannot

Table 4: Ablation study of EGA

Variant	Accuracy
Full EGA (Ours)	0.705
w/o Residual connection	0.0065
w/o Zero-initialization	0.6840
w/o L2-normalization	0.6590

Table 5: Sensitivity of different hyperparameters

Triplet Margin m	0.1	0.2	0.3	0.5
Recall@1	0.8420	0.8180	0.8160	0.7400
Recall@5	0.9782	0.9654	0.9610	0.9125
Hidden Dimension d	512	1024	2048	4096
Recall@1	0.8300	0.8340	0.8200	0.8240
Recall@5	0.9695	0.9712	0.9680	0.9688

reconstruct semantic relationships from target data alone and must operate as a refinement over the pre-trained features. ℓ_2-normalization keeps the transformed embeddings on the unit hypersphere, and removing it lowers accuracy to 0.6590. Zero-initialization lets the adapter begin as an identity mapping, and removing it lowers accuracy to 0.6840.

Table 5 evaluates stability across hyperparameter variations. EGA maintains consistent quality for margins $m \in \{0.1, 0.2, 0.3\}$, but increasing m to 0.5 degrades Recall@1 to 0.7400: a larger margin forces the adapter to update embeddings that already satisfy local neighborhood constraints, reducing gradient sparsity. Performance is also stable across hidden dimensions from 512 to 4096, indicating that EGA's behavior is determined by its structural design rather than by specific parameter choices.

4.5 Generalization to More Backbones

To verify that EGA's benefits are not specific to CLIP ViT-B/32, we further evaluate it on two stronger vision encoders: DINOv2-large [23] and SigLIP [39].

Table 6 shows consistent improvements across all three backbones. The largest gains appear on the relatively weaker CLIP ViT-B/32 ($+13.95$ pp in Label Precision@1 and $+13.20$ pp in Recall@1), but the stronger DINOv2-large and SigLIP backbones also see stable improvements ($+4.75$ pp and $+8.55$ pp in Label Precision@1 respectively). EGA's design therefore transfers to modern high-capacity vision encoders rather than relying on idiosyncrasies of the CLIP representation.

Table 6: Retrieval performance on more backbones.

Backbone	Label Precision@1		ANNS Recall@1	
	Raw	+EGA	Raw	+EGA
CLIP ViT-B/32	0.5560	**0.6955**	0.6665	**0.7985**
DINOv2-large	0.8530	**0.9005**	0.8785	**0.9310**
SigLIP SO400M	0.7740	**0.8595**	0.8015	**0.9155**

5 Conclusion

We approached adapter training from a vector search deployment perspective, where the index and query stream together cover a label space larger than the labeled subset available at adapter training time. Under this lens, different paradigms occupy distinct operating points in the (ID, OOD) Label Precision plane, and the choice among them reflects which slice of the deployed query distribution one is willing to serve poorly. EGA reaches 0.705 in-distribution Label Precision on CIFAR-100, with OOD degradation bounded at -7 to -9 pp on CIFAR-10 and Food-101 and a $+9.9$ pp improvement on FGVC-Aircraft. ICon and SRL (0.999, 0.992 in-distribution) fall 32 to 33 pp below the frozen baseline on Food-101; capacity-matched LoRA ($r = 128$) preserves OOD precision better on two of three benchmarks but trails EGA by 8.7 pp in-distribution. The benefits transfer to DINOv2-large and SigLIP (Label Precision@1 gains of 4.75 and 8.55 pp). We associate this behavior with gradient sparsity at convergence (96.5% zero-gradient triplets) and suggest it as a diagnostic for adapter designs intended to operate under unseen-class queries.

Limitations. First, our gradient-sparsity analysis (Section 3.3) is empirical rather than formal, and we do not provide a Lipschitz bound on the unseen-class perturbation. Second, EGA does not dominate every competitor on every axis: on CIFAR-10 and Food-101, capacity-matched LoRA achieves higher OOD Label Precision, and deployments dominated by OOD queries on these distributions may prefer LoRA. Third, our comparisons vary the loss family and architecture together in some cases, which prevents a clean disentanglement of the two factors. Controlled sweeps under matched conditions are the natural next step.

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A Technical Conditions for the Sparsity-Locality Claim

This appendix elaborates on the technical conditions underlying the empirical claim made in Section 3.3.

(1) Gradient support. The triplet loss in Eq. (7) can be written as $\mathcal{L} = \frac{1}{|\mathcal{B}|}\sum_{(a,p,n)\in\mathcal{B}} \ell_{apn}$, where $\ell_{apn} = \max(0, \delta_{apn} + m)$ and $\delta_{apn} = d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n)$. The subgradient with respect to θ is

$$\nabla_\theta \ell_{apn} = \mathbf{1}[\delta_{apn} + m > 0] \cdot \nabla_\theta \delta_{apn}, \tag{12}$$

which is exactly zero whenever $\delta_{apn} + m \leq 0$. Therefore, $\nabla_\theta \mathcal{L}$ receives contributions only from active triplets satisfying Eq. (8).

(2) Convergence of $\rho(t)$. Under mild regularity conditions (Lipschitz continuity of f_θ and bounded embedding norms), the sequence $\{\hat{\mathbf{z}}^{(t)}\}$ converges to a stationary point $\hat{\mathbf{z}}^*$ of $\mathcal{L}$. At stationarity, $\rho(t)$ converges to a steady-state value ρ^* determined by the within-class distance distribution of the converged embeddings, since $\rho(t)$ is a continuous functional of the embedding distribution and the embeddings converge.

(3) Empirical locality. Strictly speaking, parameter updates $\nabla_\theta \delta_{apn}$ computed on seen-class triplets affect f_θ globally (including its behavior on unseen-class inputs) since f_θ is a parametric MLP and is not locally supported on the input manifold. We therefore do not claim formal zero-gradient over unseen regions. Instead, we observe an empirical locality property: when (i) the residual structure with zero initialization keeps the cumulative perturbation magnitude bounded by the sum of training gradients, and (ii) gradient sparsity at convergence ($\rho^* \approx 3.5\%$ on FGVC-Aircraft) attenuates this sum, the resulting perturbation in unseen-class regions is small enough to preserve the pretrained semantic structure for retrieval purposes. This observation is empirically supported by the OOD Label Precision results in Section 4.2 and Figure 6, where EGA preserves or improves Label Precision on unseen categories while ICon and SRL, which lack the gradient sparsity property, collapse below the frozen baseline. A formal Lipschitz-based bound on this perturbation is left for future work.

(4) Dependence on m. The residual distance gap δ^*_{apn} at convergence is bounded by the within-class diameter $\Delta_c = \max_{x,x'\in c} d(\hat{\mathbf{z}}, \hat{\mathbf{z}}')$ for seen class c. For $m \leq \min_c \Delta_c$, the adapter can drive all within-class distances below the threshold, pushing ρ^* toward zero. For $m > \min_c \Delta_c$, some triplets remain active at convergence because the within-class spread exceeds the margin, resulting in a strictly positive ρ^*. This explains the empirically observed increase in ρ^* with larger m reported in Table 5.

B Experimental Setup Details

Backbones models. All experiments in the main paper use frozen CLIP ViT-B/32 [25] image embeddings as the base representation. To verify that EGA's benefits are not specific to this backbone, we further evaluate EGA on two stronger vision encoders: DINOv2-large [23] and SigLIP [39]. DINOv2-large is a self-supervised vision transformer trained on 142M images, while SigLIP is a supervised vision transformer trained on 400M image-text pairs. Both models produce higher-quality embeddings than CLIP ViT-B/32, as reflected in their stronger raw retrieval performance, and EGA continues to provide meaningful improvements on top of these stronger backbones, as shown in Table 6 in Section 4.4.

Adapter models. We compare EGA against the frozen CLIP baseline and four trained adapters, all sharing the same training data, evaluation protocol, and 75/25 retrieval split:

- **ICon** [1]: a global contrastive objective minimizing the KL divergence between the empirical similarity distribution and the class-membership distribution. We instantiate ICon on the same EGAMLP architecture used by our method ($\sim$4.2M parameters).
- **SRL** [4]: a structural regularization objective jointly optimizing batch-wide uniformity and within-class homogeneity, also instantiated on the EGAMLP architecture.

- **LoRA + InfoNCE** [10, 32]: a capacity-matched low-rank adapter (rank $r = 64$ and $r = 128$, $\sim$2.6M–4.2M parameters) with zero-initialized residual structure, trained with supervised InfoNCE, which is a standard global contrastive loss.
- **LoRA + Triplet** [10]: identical capacity-matched low-rank architecture as above (rank $r = 64$ and $r = 128$), but trained with the same small-margin ($m = 0.2$) triplet loss as EGA.

The two LoRA configurations isolate the effect of architectural capacity vs. loss function: comparing LoRA + InfoNCE against ICon/SRL holds the loss family roughly fixed while changing capacity, and comparing LoRA + Triplet against EGA holds the loss fixed while changing capacity.

Datasets. We evaluate on four datasets covering both in-distribution and OOD retrieval scenarios. **CIFAR-100** (100 classes, 60K images) serves as the in-distribution benchmark: the adapter is trained and evaluated on the same class label space, with a 75/25 split between database and query sets. **CIFAR-10** (10 classes, 60K images) provides a cross-dataset OOD setting: the adapter is trained on CIFAR-100 and evaluated on the disjoint CIFAR-10 classes. **FGVC-Aircraft** (100 fine-grained aircraft variants, 10K images) and **Food-101** (101 food categories, 25K images for the test split we use) provide unseen-class OOD settings: classes are randomly partitioned into 80% seen / 20% unseen with a fixed seed, the adapter is trained on the seen split, and retrieval is evaluated strictly on the unseen split. Image embeddings for all datasets are extracted with frozen CLIP ViT-B/32.

Indexing and metrics. For all retrieval evaluations we use FAISS [16] `IndexIVFFlat` with nlist$=10$ centroids and report results at nprobe $\in \{1, 5, 10\}$. We measure both Label Precision (LP@K, Eq. 1) and ANNS Recall (AR@K, Eq. 2) at $K \in \{1, 3, 5, 10\}$.

Training. All adapters are trained for 150 epochs with AdamW (weight decay 10^{-4}), cosine-annealed learning rate, and batch size 256. EGAMLP-based methods (EGA, ICon, SRL) use learning rate 10^{-4}; LoRA variants use 10^{-3} to compensate for their smaller parameter count. Triplet-based methods sample one positive and one negative per anchor uniformly within each mini-batch; global contrastive methods (ICon, SRL, LoRA + InfoNCE) operate on the full pairwise similarity matrix. All experiments use a fixed random seed (42) for reproducibility.

Compute platform. All experiments were conducted on the Chameleon Cloud [17] platform using a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA A100 GPU of 80 GB HBM2e. The operating system is Ubuntu 24.04 LTS.

C Additional Experimental Results

Robustness across search budgets. Figure 6 traces Label Precision@1 across three OOD benchmarks and three search budgets (nprobe $\in \{1, 5, 10\}$). The collapse pattern of ICon and SRL persists across all 18 measurement points (3 datasets $\times$ 3 budgets $\times$ 2 methods), ruling out the possibility that the degradation is an artifact of any particular benchmark or indexing configuration. EGA's relative ordering against other methods is also stable across budgets.

Linear Probing Baseline Analysis To address potential concerns about additional baselines, we also evaluated Linear Probing as an example. As shown in Table 7 and Table 8, Linear Probing achieves reasonable in-distribution performance (0.5112 LP@1 on CIFAR-100) but completely collapses on out-of-distribution benchmarks (0.0065–0.0133 LP@1). This confirms that Linear Probing is not a meaningful comparison in our cross-dataset OOD setting due to class semantic mismatch.

Table 7: Linear Probing in-distribution performance on CIFAR-100 (75/25 split, mean $\pm$ std over 3 seeds).

Method	LP@1	AR@1
Linear Probing	0.5112 $\pm$ 0.0020	0.7533 $\pm$ 0.0025

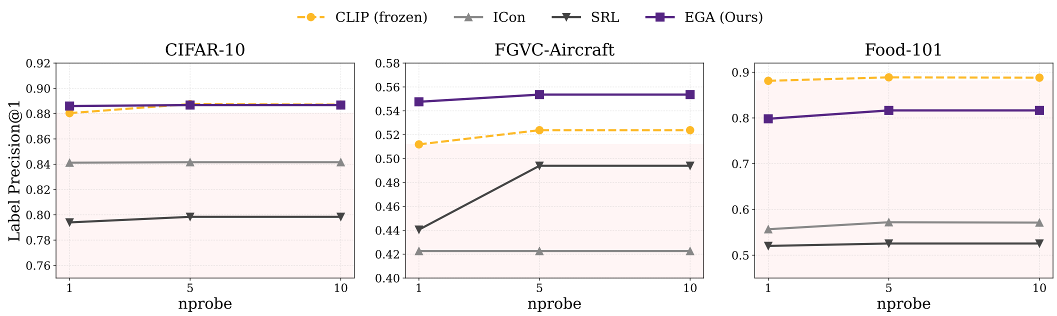

Figure 6: Label Precision@1 across three OOD benchmarks and three search budgets (nprobe $\in \{1, 5, 10\}$).

Table 8: Linear Probing OOD performance (mean $\pm$ std over 3 seeds).

Dataset	LP@1	AR@1
Aircraft	0.0133 ± 0.0038	0.4401 ± 0.0254
Food-101	0.0077 ± 0.0019	0.5162 ± 0.0228
CIFAR-10	0.0065 ± 0.0005	0.6454 ± 0.0096